

Using Social Media Analytics to Improve Artist Popularity: A Case Study within Ed Sheeran

1. INTRODUCTION

Edward Sheeran is an English singer-songwriter. He began recording music in 2004. He is still active and will tour Australia & New Zealand tour on Feb 2023 (Ed Sheeran Australia Tour, n.d.). "The discography of Ed Sheeran consists of four solo studio albums, one collaboration album, seventeen extended plays, one video album, forty-three singles, including seventeen singles as a featured artist, seven promotional singles, one box set, and 71 music videos, 12 of which have been UK number-ones." (Wikipedia Contributors, 2019, p.1).

The meaning behind Ed Sheeran's song lyrics is realistic and touches the heart. For example, his song 'You Need Me, I Don't Need You' "takes a shot at other artists who fail to write their music or want to find success without putting in the hard work first" (Ed Sheeran - "the a Team," n.d. p11).

Ed supports many charities which typically aimed at helping disadvantaged and vulnerable individuals. Such as homelessness, children with life-threatening illnesses, and sex workers (Does Ed Sheeran Give to Charity? n.d.). Recently, Ed showed in the Concert for Ukraine, which is a benefit concert to raise vital funds for the humanitarian needs of the people of Ukraine (Ed Sheeran, Camila Cabello and Emeli Sandé to Play Concert for Ukraine, 2022).

2. PURPOSE

As David Andrew Wiebe Said, “A successful music career always begins with great music. But these days, you also need to focus your efforts on promoting your music. Otherwise, no one will hear it!” (Wiebe, 2018, P6). The first thing is to go where listeners are and plan to promote your music. Social media can be helpful. In this case study, we will be using the platforms Twitter, Spotify, and YouTube to improve artist Edward Sheeran’s popularity.

Our goal is to improve artist popularity. We can select related topics and keywords to get data set using social media, and then the data can be analyzed and reported through visualizations. They can help us:

- spot trends related to offerings and brands,
- understand conversations — what is being said and how it is being received.
- derive customer sentiment towards products and services.
- gauge response to social media and other communications.
- identify high-value features for a product or service.
- uncover what competitors are saying and its effectiveness.
- map how third-party partners and channels may affect performance (IBM, n.d.).

Twitter can analyze the trends of related topics, the characteristics of fans base and their network. YouTube and Spotify can be used to analyze audio features for all songs by an artist and the relationship between artists. Any related data of the artist could be considered. From audience demographics to traffic sources, right down to the keywords people use to find your videos. (Newberry, Sehl, 2018). Through those social media, we can know what topics are popular now, the characteristics of fans and so on. We hope social media analytics can help Ed Sheeran’s company better understand the needs and expectations of his fans and better develop marketing strategies.

3. DATA SELECTION & EXPLORATION

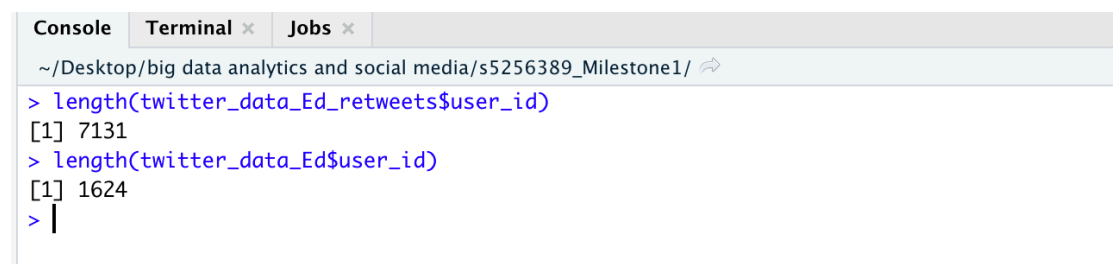
3.1 Collect data about your artist/band from Twitter. Make sure to choose keywords for data retrieval that are most relevant to your artist/band. However, try not to be too narrow. As a rough guide, you should retrieve at least 1000 tweets. List the keywords and explain what you have done and how much data you have collected.

While collecting data about an artist from Twitter, keywords (words, phrases, hashtags) chosen are important. I would like to know:

1. which Twitter users tagged EdSheeran (@EdSheeran)
2. all content related to Ed Sheeran (#EdSheeran or Ed Sheeran)
3. the popularity of the new album (2step)
4. Twitter users' engagement with upcoming concert(s) (#MathematicsTour)
5. the characteristics of Ed Sheeran's popular album or song (No.6 Collaborations Project) (the shape of you)

In the beginning, I would like to know all content related to Ed Sheeran. We use keyword #EdSheeran to collect data and then research further.

- 1) Build file ED_Sheeran.R file on RStudio
- 2) Choose the keyword “#EdSheeran” to retrieve some recent tweets. Save as data file named “TwitterData_Retweets.rds” (included retweets) and “TwitterData.rds”, respectively.
- 3) Files “TwitterActor_Ed.graphml”, “TwitterSemantic_Ed.graphml” and “TwitterSemanticMoreTerms_Ed.graphml” are also saved for further study in the platform Gephi. See the content of file “ED_Sheeran.R” for detail.
- 4) There are 7131 user records (including retweets) and 1624 user records (just tweets)



```
Console Terminal x Jobs x
~/Desktop/big data analytics and social media/s5256389_Milestone1/
> length(twitter_data_Ed_retweets$user_id)
[1] 7131
> length(twitter_data_Ed$user_id)
[1] 1624
> |
```

3.2 List the top 5 most influential users for your artist/band. Find out what other interests/characteristics they have besides those related to your artist/band. Do these 5 have something in common?

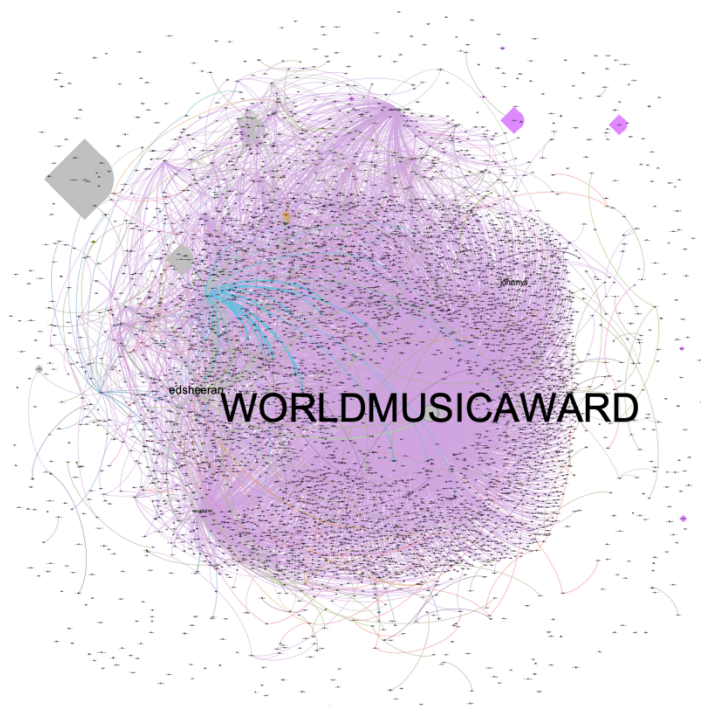
We use the PageRank function in platforms RStudio and Gephi, respectively. We can find the top 5 users are the same but in a different order on the two platforms. The reason is that the PageRank algorithm is non-deterministic. The screenshot is below:

```
> head(rank_twitter_actor_Ed_retweets, n = 5)
WORLDMUSICAWARD      wooodoojinn      edsheeran      johnnys winterVerry1230
0.248186704      0.131015112      0.083892001      0.014111934      0.009803412
```


screen_name	PageRank ▼
WORLDMUSICAWARD	0.109951
edsheeran	0.043076
johnnys	0.031197
woooojinn	0.017847
winterVerry1230	0.008961

We searched the screen name on Twitter and the browser engine as below:

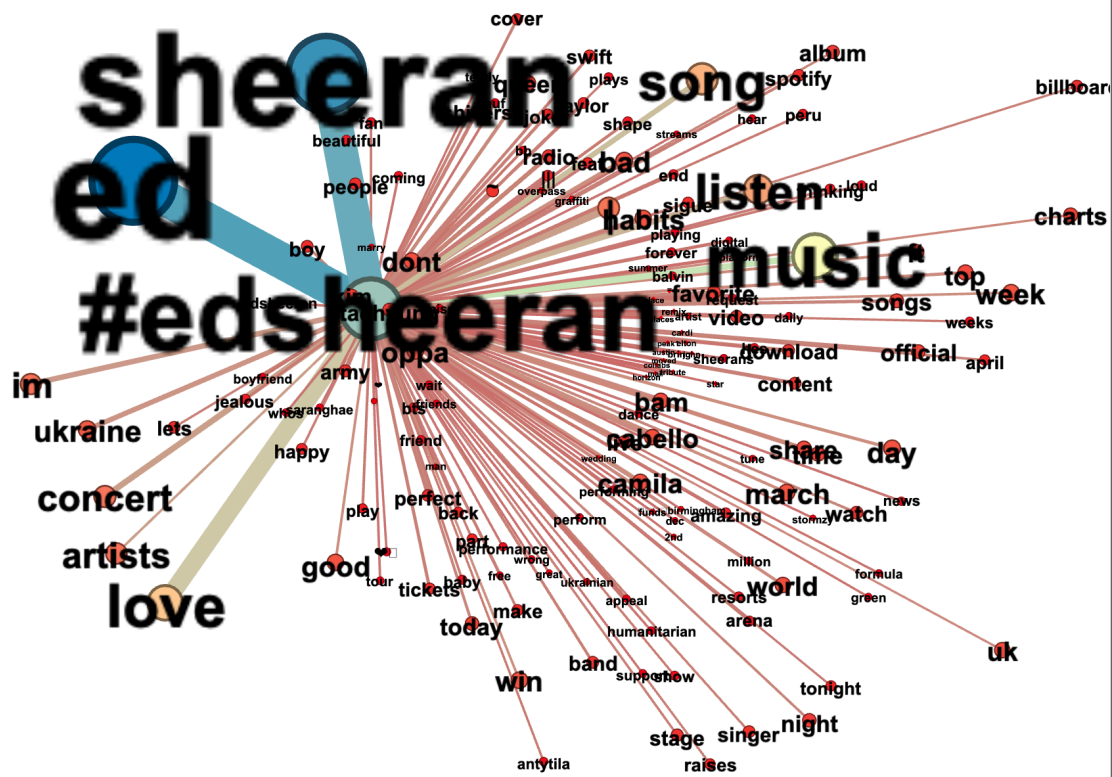
- The user Edsheeran is the artist we want to analyze.
- The user WORLDMUSICAWARD is an international award presented to the world's best-selling artists in several categories and the best-selling artists from each significant territory.
- The user Johnnys is a Japanese agency, which trains and promotes groups of male entertainers known as Johnny's
- The user Woooojinn and winterVerry1230 are Korean singers



According to the above Gephi Force Atlas Layout, we can find that.

1. Ed Sheeran and other singers and agents have their own communities.
2. Ed Sheeran, winter Verry 1230 and WORLDMUSICAWARD belong to one community.

3.3 List the top 10 most important terms that appear together with your keyword(s) related to your artist/band. Explain the results.



name	After analysing the file TwitterSemantic_Ed.graphml in Gephi, we got the left top 10 terms related to the keyword #edsheeran. Combining the above screenshot, let us try to explain it.
ed	
sheeran	
music	
love	
song	
listen	
concert	
artists	
im	

1. Looking at the above graph, we can quickly notice the phrase “Ukraine” and “concert”. Then we searched on Twitter and the Brower engine. We know it is Ukraine Benefit Concert that just happened recently.
2. The word “artist”, “song”, “listen”, and “music” are related to music. That’s why they appeared many times.
3. We also notice the word “love”, “good”, “bad”, and “perfect” that may indicate that users actively participated in evaluating the quality of music. The dataset maybe is suitable for further sentiment analysis.
4. The word “IM” may be an abbreviation or some new buzzword. We can study it further.

3.4 Calculate how many of your retrieved tweets are retweets. Alternatively, if you filtered out retweets in your query, calculate how many unique user accounts there are in your dataset. Explain the code you have used for the calculation. What do the results tell you?

We use the collect function and set property retweets = TRUE or FALSE to get our dataset TwitterData_Retweets or TwitterData, respectively. There are 5507 retweets and 863 unique users posted tweets. The result tells us that one tweet has averaged about six retweets. Tweets indicate how prolific a user is, and retweets tell us how willing people are to amplify your messages and help them spread.

```
~/Desktop/big data analytics and social media/s5256389_Milestone1/    
> length(twitter_data_Ed_retweets$user_id)  
[1] 7131  
> length(twitter_data_Ed$user_id)  
[1] 1624  
> length(twitter_data_Ed_retweets$user_id)-length(twitter_data_Ed$user_id)  
[1] 5507  
> length(unique(twitter_data_Ed$user_id))  
[1] 863  
> 5507/863  
[1] 6.381228  
> |
```

4. DATA SELECTION & EXPLORATION(CONTINUED)

4.1 Use the Spotify API to extract data about your artist/band. For example:

1) How many albums & songs have they published?

According to Spotify, there are some songs in different albums.

To summarize, they have published 12 albums and 138 songs (Note: new songs are released on 26 May 2022, some data haven't been updated immediately)

Detailed as below:

```
# Retrieve album data of artist
albums <- getAlbums("6eUKZXaKkcviH0Ku9w2n3V", token = keys)
View(albums)

> unique(albums$name)
[1] "= (Tour Edition)"          "2step (The Remixes)"      "="
[4] "No.6 Collaborations Project" "÷ (Deluxe)"               "5"
[7] "x (Deluxe Edition)"        "x (Wembley Edition)"      "Live at the Bedford"
[10] "Loose Change"              "+"                          "No.5 Collaborations Project"

# Retrieve songs from all albums
album_ids <- getAlbums("6eUKZXaKkcviH0Ku9w2n3V", token = keys) %>% select(id)

for (i in 1:nrow(album_ids))
{
  album_songs <- getAlbum(sapply(album_ids,'[[',i), token = keys) %>% select(id,name)
  if(i==1) {
    all_songs=album_songs
  } else {
    all_songs=bind_rows(all_songs,album_songs)
  }
}
View(all_songs)
```

There are 199 songs in all albums. But 69 songs are duplicated.

```
> nrow(all_songs)
[1] 199
> sum(duplicated(all_songs$name))
[1] 69
> length(unique(all_songs$name))
[1] 130

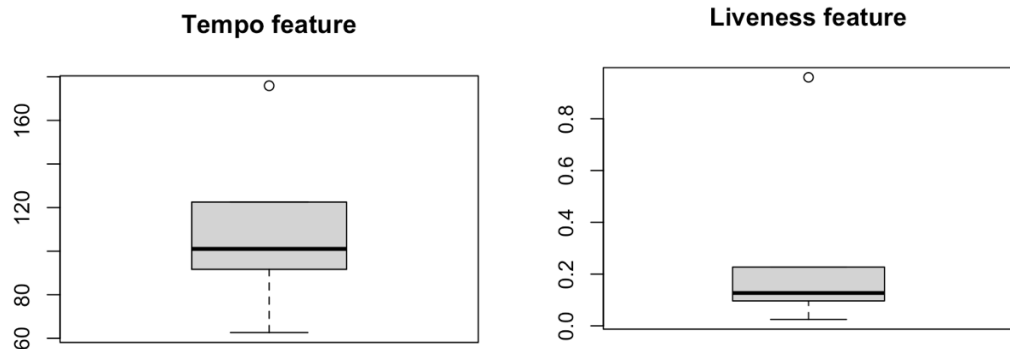
> length(unique(audio_features$track_name))
[1] 138
```

2) What are the prevalent features of their songs (e.g., valence)?

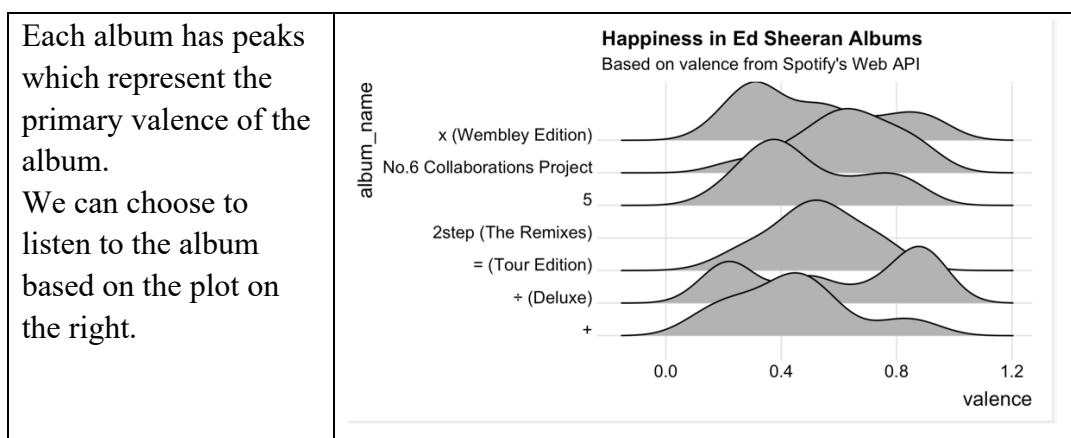
- a) We will analyse Ed Sheeran's audio's tempo and liveness using statistics five mums: Tukey min, lower-hinge, median, upper-hinge, max. as the below we can see.

```
> fivenum(audio_features$tempo)
[1] 62.631 91.670 101.042 122.531 175.908
> fivenum(audio_features$liveness)
[1] 0.0248 0.0964 0.1270 0.2270 0.9600
~
```

- Tempo: Tempo ranges from 62.631 to 175.908. The average is 101.042. The overall Tempo of Ed Sheeran is moderately fast. He also has some relatively slow or extremely fast audios.
- Liveness: Liveness ranges from 0.0248 to 0.9600, and the average is 0.1270. The overall Liveness is low. It means that most songs are recorded privately. Some of the songs were recorded in public (concert)



- b) We plot happiness valence scores for each album as below.



3) With whom have they often collaborated?

```
# Retrieve information about related artists
```

```
related_bm <- getRelated("Ed Sheeran", token = keys)
View(related_bm)
```

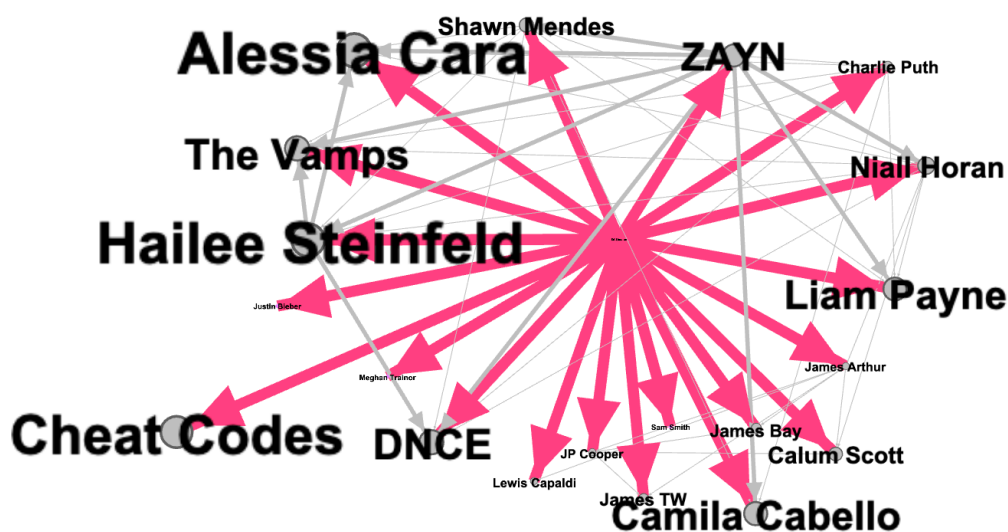
	name	id	popularity	type	followers
1	Shawn Mendes	7n2wHs1TKAczGzO7Dd2rGr	84	artist	37857653
2	James Arthur	4IWBUAFlprNtaOHcjPRM	78	artist	9794821
3	James TW	0B3N0ZINFWvizfa8bKiz4v	66	artist	1108268

Showing 1 to 3 of 20 entries, 5 total columns

Get the top 100 artists and convert network to graph `RelateArtists.graphml` and then visualize it in Gephi. We can find Ed Sheeran often collaborate with Camila Cabello, Liam Payne, Alessia Cara, Shawn Mendes and so on.

```
# Create a network of artists related to the Top 100 artists
topsongs <- getPlaylistSongs("spotify", "4h0KQuZbraPDIfaGbM3lKI", token = keys)

edges <- c()
for (artist in topsongs$artist){
  related <- getRelated(artist, token = keys)
  for (relatedartist in related$name){
    edges <- append(edges, artist)
    edges <- append(edges, relatedartist)
  }
}
```



4) How many years have they been active? And how does the Spotify data compare to the information you collected from other sources in Step 3.1?

```
> unique(audio_features$album_release_year)
[1] 2022 2019 2017 2015 2014 2013 2011
```

	Active year	Albums & songs
Search from other sources	2004-now	The discography of Ed Sheeran consists of four solo studio albums, one collaboration album, seventeen extended plays, one video album, forty-three singles,
Spotify	2011-now	12 Albums and 138 Songs
Conclusion	Spotify was founded on 2016. Ed Sheeran began to use Spotify in 2011.	We can use analytics from Spotify for Artists to get a more granular look at the demographics of an artist's audience. The platform displays information such as their age, gender, and the country and city they live in. Using these analytics will better assist market artists. We need to search on browser engines, books, etc., for more text information.

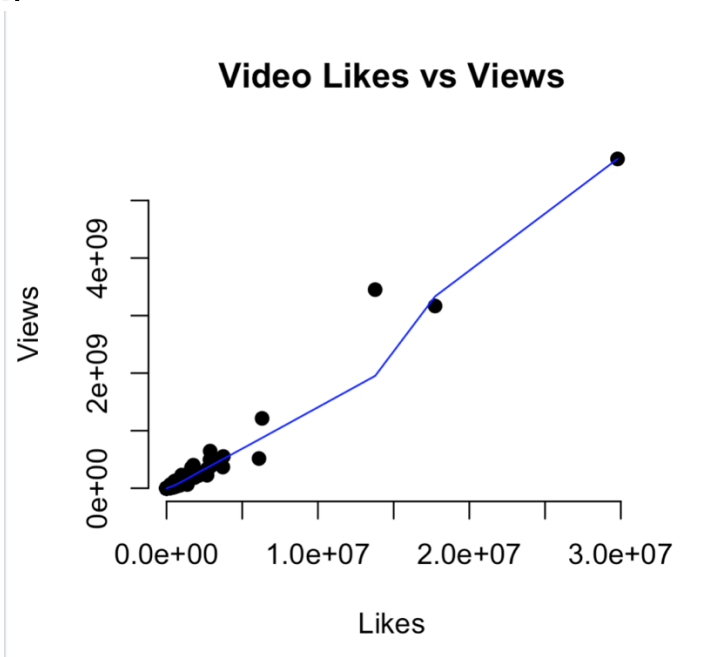
4.2 Retrieve data relevant to your artist/band from YouTube. Which videos have the highest number of views and likes? Do you see a correlation between views and likes? (Your dataset may contain hundreds of videos, so it's OK if you choose only a subset of those to get their statistics, in order to avoid hitting the rate-limit. However, you should get statistics for at least 5 videos.) (=> Lab 3.2)

- 1) **Highest number of likes:** Don't [Official Music Video] and **Highest number of views:** "Help Falling In love (Elvis All Star Tribute 2019)"

```
> #highest number of likes.
> video_search[video_search$video_id == videoid_max_likeCount$id,c("video_id","title")]
      video_id                                     title
83 iD2rhdfRehU Ed Sheeran - Don't [Official Music Video]
> #highest number of views
> video_search[video_search$video_id == videoid_max_viewCount$id,c("video_id","title")]
      video_id                                     title
66 GNd0afTpls Ed Sheeran - Can't Help Falling In Love (Elvis All Star Tribute 2019)
```

- 2) The following scatter plot shows that views and likes are a positive correlation, and most of the Views and Likes are around 177396 and 14992692, respectively.

```
# plot likeCount and viewCount scatter plot
plot(stats$likeCount,stats$viewCount,main="Video Likes vs Views",xlab = "Likes", ylab =
"Views", pch = 19, frame = FALSE)
lines(lowess(stats$likeCount, stats$viewCount), col = "blue")
```



```
> fivenum(as.numeric(stats$likeCount))
[1]      0.0      5410.0     177396.0    765597.5   29787163.0
> fivenum(as.numeric(stats$viewCount))
[1]      60      444124     14992692    85433203   5722218763
~ |
```


5. TEXT PRE-PROCESSING

5.1 Perform text pre-processing and create a Term-Document Matrix for your Twitter data. What are the 10 terms occurring with the highest frequency? How are they different to your answer for 1.5) above? (=> Lab 2.2)

Import “TwitterData_Retreets.rds”, Term-Document Matrix as below:

```
> view(inspect(doc_term_matrix))
<<DocumentTermMatrix (documents: 7131, terms: 2772)>>
Non-/sparse entries: 95676/19671456
Sparsity : 100%
Maximal term length: 36
Weighting : term frequency (tf)
Sample :
  Terms
Docs  america arrow eaabedefu eashiv globe keycap music note show top
2000  0      0      0      0      0      0      0      0      0      0
279   0      0      0      0      0      0      0      0      0      0
304   1      1      0      0      1      0      3      3      1      0
462   0      0      0      0      0      0      0      0      0      0
532   0      0      0      0      0      0      0      0      0      0
6266  0      0      0      0      0      0      0      0      0      0
889   0      0      0      0      0      0      0      0      0      0
895   0      0      0      0      0      0      0      0      0      0
90    0      0      0      0      0      0      0      0      0      0
989   0      0      0      0      0      0      0      0      0      0
```

After importing the previous dataset TwitterData_Retreets.rds. The following 10 terms occurring with the highest frequency as below:

PageRank:

```
head(rank_twitter_semantic, n = 10)
#edsheeran #camilacabello #eltonjohn #dualipa ed
0.08899732 0.04218823 0.02970633 0.02850860 0.02067871
sheeran 📈 10 🎵 🌐 🎵 #jacksonwang 📺 bambam #shivers
0.01908220 0.01533265 0.01517977 0.01325144 0.01283967
```

Term-Document (Raw text)

```
> head(freq1, n = 10)
#edsheeran #camilacabello #dualipa #eltonjohn #gayle #avamax
10310 3568 2955 2952 2935 2922
#tiesto 📈 10 🎵 🌐 🎵 📺 bambam #mariahcarey
2850 2705 2275 2216
```

Term-Document (Clean text)

```
> head(freq, n = 10)
keycap music note show america globe top arrow eashiv
6473 6174 6049 3736 3627 3626 3487 3429 2932
eaabedefu
2928
```

Using raw text, getting the highest frequencies of words by using the PageRank algorithm and Term Document algorithm is similar. Using clean text can get a different result because clean text removes hashtags, emoticons, etc.

6. SOCIAL NETWORK ANALYSIS

6.1 For your

- Twitter actor graph,
- Twitter semantic graph, and
- YouTube actor graph,

explain each of its properties and attributes (highlighted in the sample screenshot below).

<pre>> summary(yt_actor_graph) IGRAPH 938a672 DN-- 80026 101472 -- + attr: type (g/c), name (v/c), screen_name (v/c), node_type (v/c), label (v/c), video_id (e/c), comment_id (e/c), edge_type (e/c)</pre>												
In yt_actor_graph, there are 80026 nodes and 101472 edges.												
g/c: graph cover												
v/c: vertex cover												
e/c: edges cover												
Graph type attribute:	Directed or Undirected or mix											
Nodes attributes:	<table><tr><td>name</td><td>screen_name</td><td>node_type</td><td>Label</td></tr><tr><td>UCPab1PdjkWjqDGBzq75wF8Q</td><td>HiTtaMaNeefe'GBE</td><td>actor</td><td>HiTtaMaNeefe'GBE (...)</td></tr></table>				name	screen_name	node_type	Label	UCPab1PdjkWjqDGBzq75wF8Q	HiTtaMaNeefe'GBE	actor	HiTtaMaNeefe'GBE (...)
name	screen_name	node_type	Label									
UCPab1PdjkWjqDGBzq75wF8Q	HiTtaMaNeefe'GBE	actor	HiTtaMaNeefe'GBE (...)									
Edges attributes:	<table><tr><td>video_id</td><td>comment_id</td><td>edge_type</td></tr><tr><td>.. orJSJGHjBLI</td><td>Ugzn-0s7dd6gR...</td><td>comment</td></tr></table>				video_id	comment_id	edge_type	.. orJSJGHjBLI	Ugzn-0s7dd6gR...	comment		
video_id	comment_id	edge_type										
.. orJSJGHjBLI	Ugzn-0s7dd6gR...	comment										

<pre>> summary(twitter_actor_graph) IGRAPH f5556a7 DN-- 5470 7490 -- + attr: type (g/c), name (v/c), screen_name (v/c), label (v/c), edge_type (e/c), timestamp (e/c), status_id (e/c)</pre>									
In twitter_actor_graph, there are 85470 nodes and 7490 edges.									
g/c: graph cover									
v/c: vertex cover									
e/c: edges cover									
Graph type attribute:	Directed or Undirected or mix								
Nodes attributes:	<table><tr><th>name</th><th>screen_name</th><th>Label</th></tr><tr><td>273461792</td><td>EventFriend</td><td>EventFriend (273461792)</td></tr></table>			name	screen_name	Label	273461792	EventFriend	EventFriend (273461792)
name	screen_name	Label							
273461792	EventFriend	EventFriend (273461792)							
Edges attributes:	<table><tr><th>edge_type</th><th>timestamp</th><th>status_id</th></tr><tr><td>retweet</td><td>2022-04-04 10...</td><td>15109339383612...</td></tr></table>			edge_type	timestamp	status_id	retweet	2022-04-04 10...	15109339383612...
edge_type	timestamp	status_id							
retweet	2022-04-04 10...	15109339383612...							

<pre>> summary(yt_actor_graph) IGRAPH 71aab73 DN-- 3363 4419 -- + attr: type (g/c), name (v/c), screen_name (v/c), node_type (v/c), label (v/c), video_id (e/c), comment_id (e/c), edge_type (e/c)</pre>												
In twitter_actor_graph, there are 3363 nodes and 4419 edges. g/c: graph cover v/c: vertex cover e/n: edge to node												
Graph type attribute:	Directed or Undirected or mix											
Nodes attributes:	<table><thead><tr><th>name</th><th>screen_name</th><th>node_type</th><th>Label</th></tr></thead><tbody><tr><td>UCzggR-cvWwPAZnCbufKPmYQ</td><td>CAMA DANIELE</td><td>actor</td><td>CAMA DANIELE (UCzggR-cvWwPAZnCbufKPm...</td></tr></tbody></table>				name	screen_name	node_type	Label	UCzggR-cvWwPAZnCbufKPmYQ	CAMA DANIELE	actor	CAMA DANIELE (UCzggR-cvWwPAZnCbufKPm...
name	screen_name	node_type	Label									
UCzggR-cvWwPAZnCbufKPmYQ	CAMA DANIELE	actor	CAMA DANIELE (UCzggR-cvWwPAZnCbufKPm...									
Edges attributes:	<table><thead><tr><th>video_id</th><th>comment_id</th><th>edge_type</th></tr></thead><tbody><tr><td>orJSJGHjBLI</td><td>UgylZ-AzEshJfPqkos...</td><td>comment</td></tr></tbody></table>				video_id	comment_id	edge_type	orJSJGHjBLI	UgylZ-AzEshJfPqkos...	comment		
video_id	comment_id	edge_type										
orJSJGHjBLI	UgylZ-AzEshJfPqkos...	comment										

6.2 Perform centrality analysis by detecting degree centrality, betweenness centrality, and closeness centrality. Explain how relevant the results are to your artist/band. What is the actual degree, betweenness, and closeness centrality scores for your artist/band node in the network? Compare these scores to the scores for related artists. (=> Lab 3.1)

According to previous related Artist analysts, we will analyse the centrality analysis of related artists Camila Cabello, and Shawn Mende

Import data TwitterData_Retreets.rds and get TwitterTwomode.graphml

Id	name	Closeness Centrality	Betweenness Centr...	Degree	In-Degree	Eccentricity
n27	#shawnmendes	0.362606	467183.441179	1823	1823	8.0
Id	name	Closeness Centrality	Betweenness Centr...	Degree	In-Degree	Eccentricity
n30	@edsheeran	0.34906	4004051.375036	745	745	6.0
Id	name	Closeness Centrality	Betweenness Centr...	Degree	In-Degree	Eccentricity
n16	#camilacabello	0.442217	3921807.625739	2822	2822	6.0

Import data YoutubeData.rds and get youtubeActor_Ed.graphml

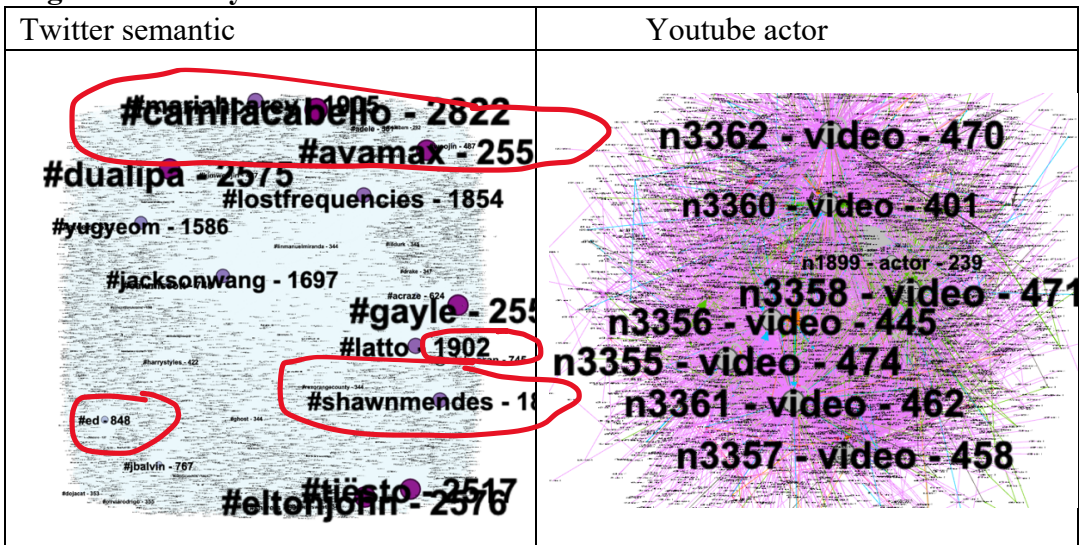
Id	screen_name	Degree	In-Degree	Out-Degree	Closeness Centrality		Betweenness Centrality	Eccentricity
n522	Ed Sheeran	20	19	1	0.285666		31878.010391	6.0

Id	screen_name	node_type	In-Deg...	Out-Degree	Degree	Closeness Centrality		Betweenness Centrality	Eccentricity
n1899	NEW TALENT	actor	0	239	239	0.286542		167354.935943	6.0

Id	screen_name	node_type	In-Deg...	Out-Degree	Degree	Closeness Centrality		Betweenness Centrality	Eccentricity
n3360	NA	video	400	1	401	0.285012		1413502.806624	6.0

Id	screen_name	node_type	In-Deg...	Out-Degree	Degree	Closeness Centrality		Betweenness Centrality	Eccentricity
n3362	NA	video	469	1	470	0.327648		1491704.443126	6.0

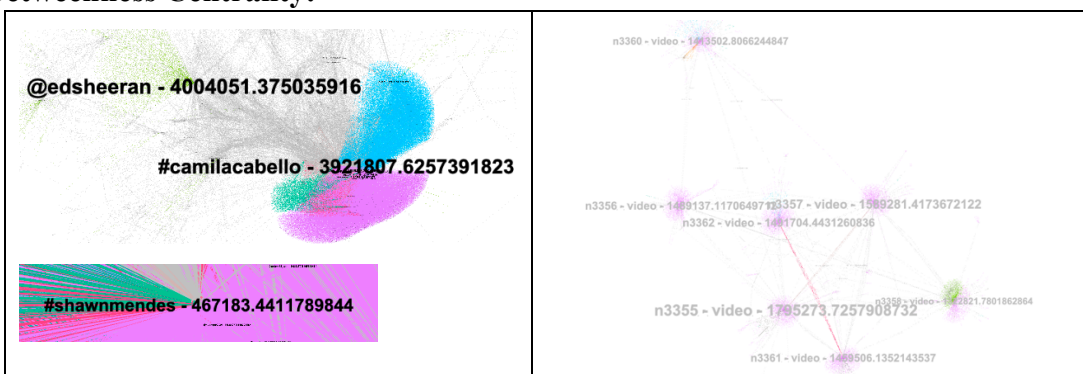
Degree centrality



Conclusion:

- #CamilaCabello degree is 2822
- #ShawnMendes degree is 1823
- #ed degree is 848
- @edsheeran degree is 745
- Artists haven't followed any other users because there is no out-degree.
- Ed Sheeran related artists have more followers than himself.
- Ed Sheeran's degree is 20
- NEW TALENT degree is 239
- videos on the above degree are around 450.

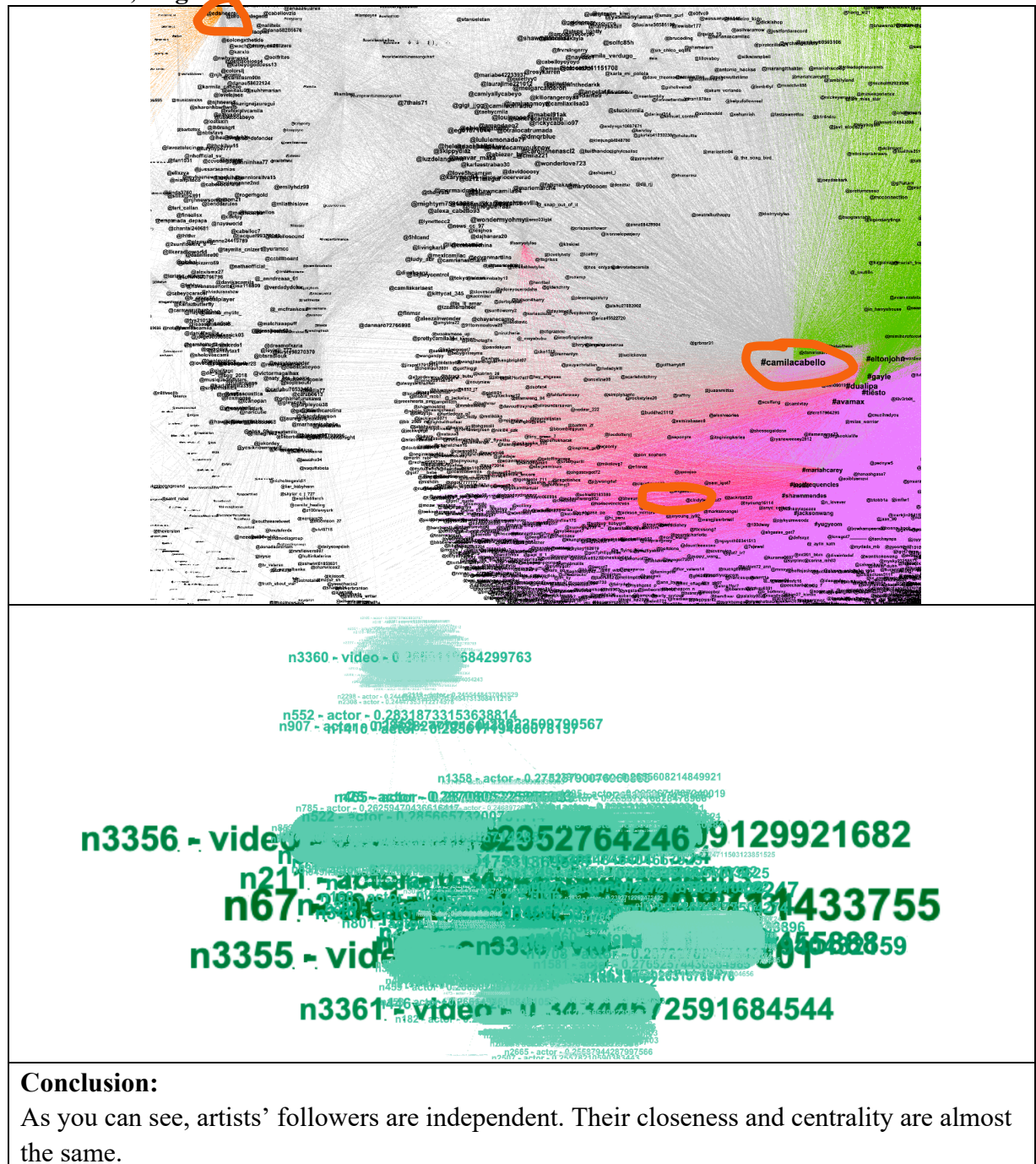
Betweenness Centrality:



Conclusion:

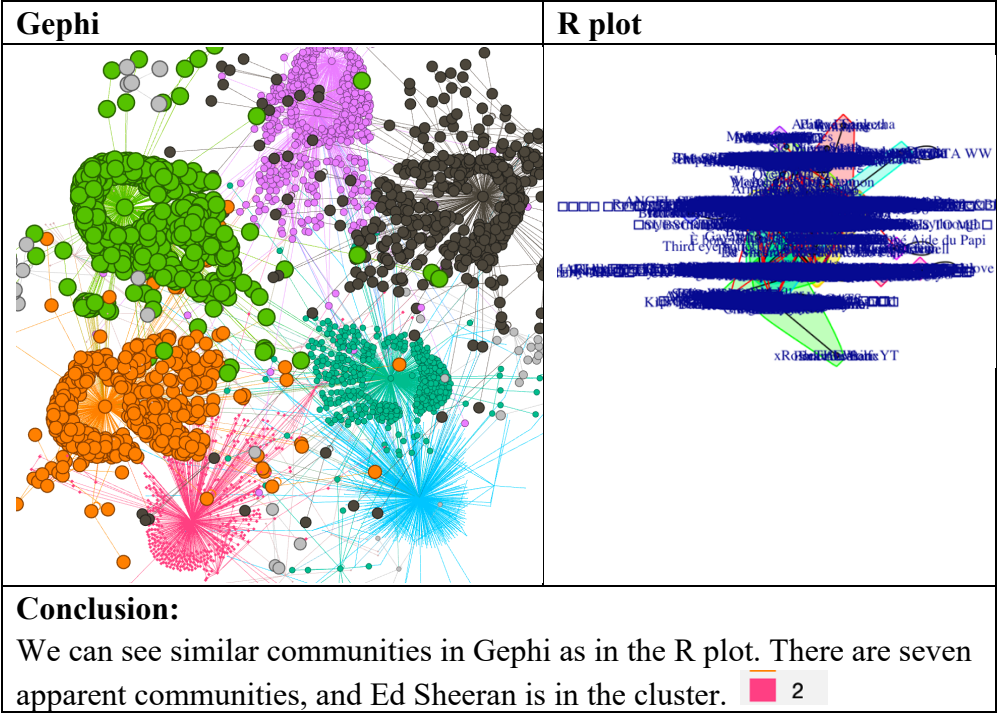
- Node @edsheeran has the highest betweenness centrality in the network
- Videos which have high degrees also have high betweenness centrality
- Artists usually have high betweenness centrality

In Rstudio, we got that the artists in and out closeness are 1.

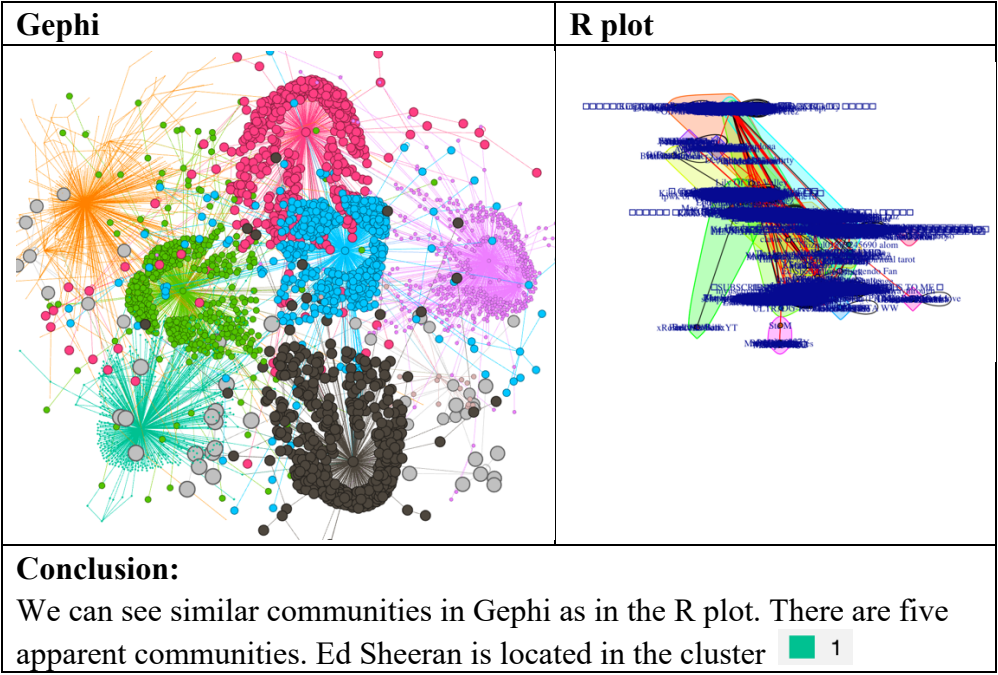


6.2 Perform community analysis with the Girvan-Newman (edge betweenness) and Louvain methods. Explain how relevant the results are to your artist/band. Perform the community analysis also for related artists. Is their community structure similar? (=> Lab 3.2)

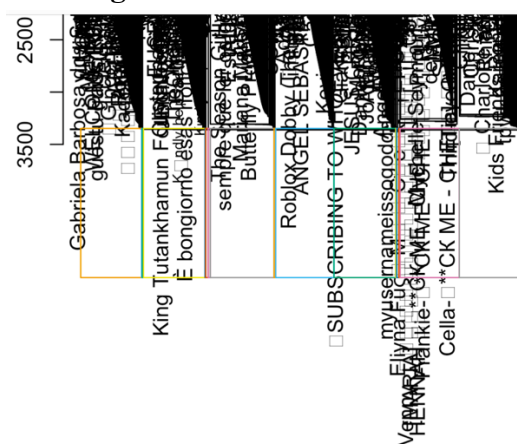
Louvain-youtube



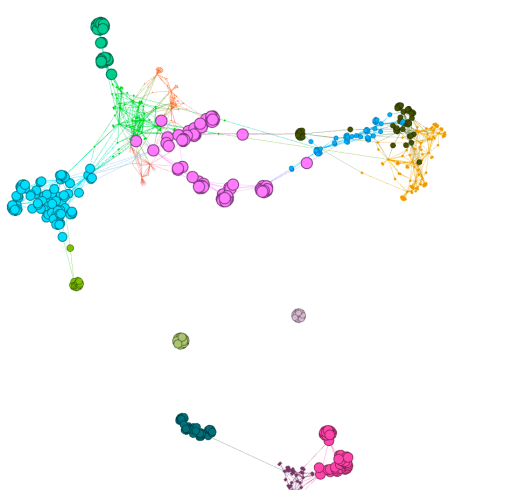
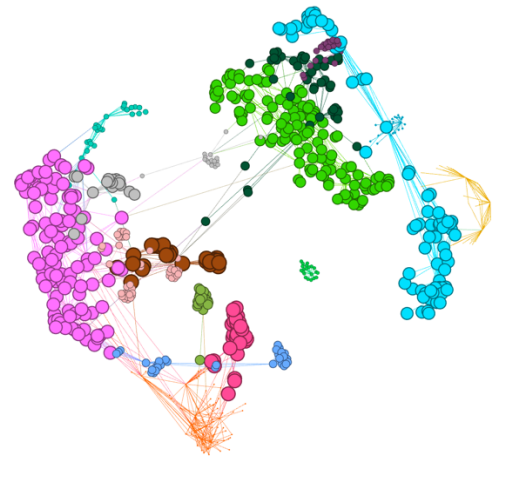
Girvan-Newman (edge betweenness)-youtube



Dendrogram



Spotify Related Artists

Louvain	Girvan-Newman (edge betweenness)
	
Shawn Mendes, and Camila Cabello are in class 2, and Ed Sheeran is in class 0	Ed Sheeran, Shawn Mendes, and Camila Cabello are in class 14
Conclusion: We can see different communities by using different methods. We prefer to use communities by Girvan-Newman (edge betweenness) because related artists are in the same communities.	

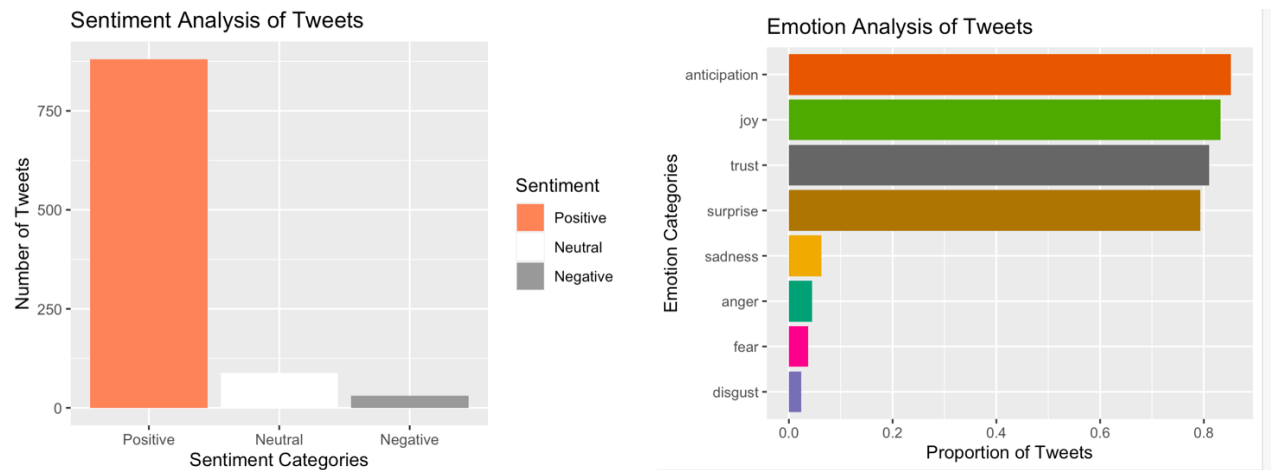
When analysing YouTube data, we get a similar community structure by using Louvain and Girvan-Newman (edge betweenness).

When analysing Spotify-related artist data, we get an entirely different community structure by using Louvain and Girvan-Newman (edge betweenness).

7. MACHINE LEARNING MODELS

7.1 Use sentiment analysis to identify how the public reacts to events and/or topics related to your artist/band. Provide a summary of public opinions (emotions, reactions). (=> Lab 5.2)

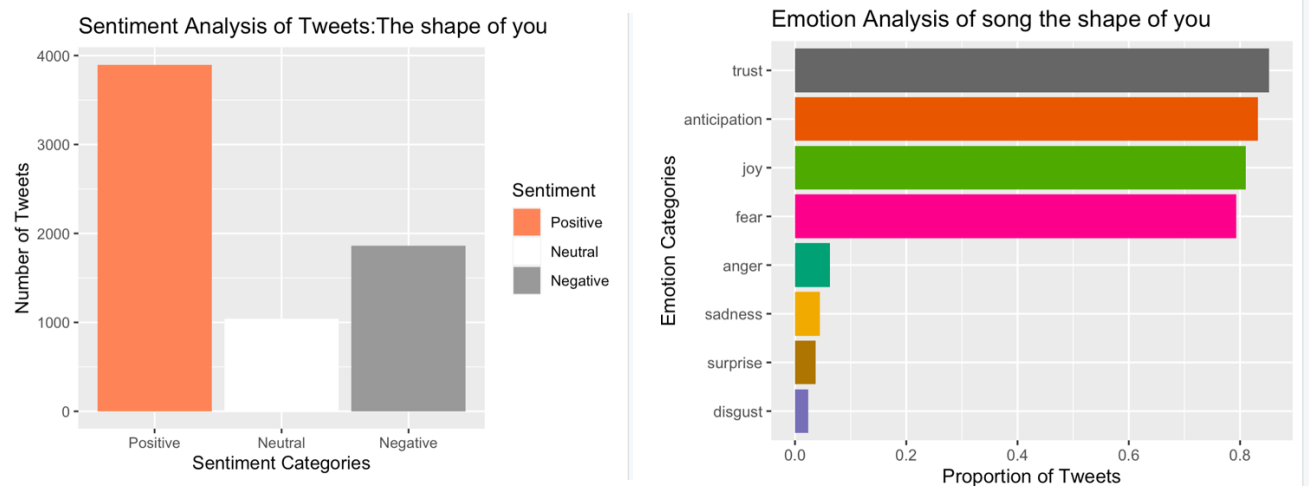
Sentiment analysis of Tweets based on the actor



Conclusion:

1. Most listeners have positive reviews on Ed Sheeran and very few negative reviews.
2. The top 4 emotions by listeners are anticipation, joy, trust, and surprise.

Sentiment analysis of Tweets based on the song "the shape of you."



Conclusion:

1. Many listeners have positive comments on the song "Shape of You", but many negative comments are still.
2. The top 4 emotions by listeners are trust, anticipation, joy, and fear.

7.2 Build a decision tree and evaluate its performance in predicting whether a song is by your artist/band. (=> Lab 5.2)

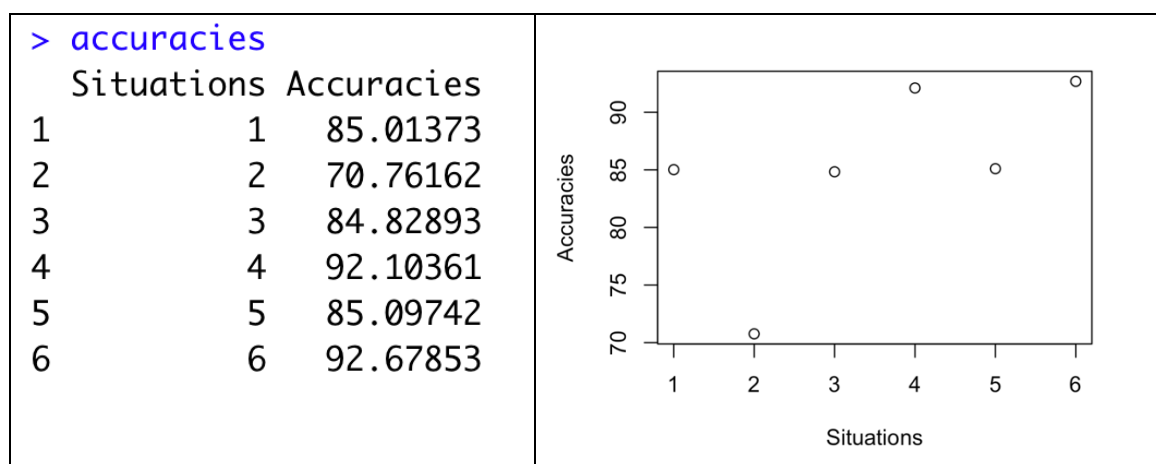
We get a dataset from features of Ed Sheeran, Shawn Mendes and Top100 songs.
The samples as below

```
> nrow(ShawnMendes_features)
[1] 208
> nrow(EdSheeran_features)
[1] 599
> nrow(top100_features)
[1] 100
```

Situation 1 Samples:(100+599) =699	Situation 2 Samples: (208+599) =807	Situation 3 Samples: (100+208+599) =907
top100 features (1) EdSheeran features (1)	Shawn Mendes features (1) EdSheeran features (1)	top100 features, (1) EdSheeran features (1) Shawn Mendes features (1)
Situation4 Samples:(100+599*2) =1298	Situation 5 Samples:(208+599*2) =1406	Situation 6 Samples: (100+208+599*2) =1506
top100 features (1) EdSheeran features (2)	Shawn Mendes features (1) EdSheeran features (2)	top100 features, (1) EdSheeran features (1) Shawn Mendes features (2)

We can see from the below that:

- There are more datasets; the model becomes more accurate.
- Comparing situation 2 and situation 5, the accuracy has a significant improvement. Although the two singers' styles are similar, you can still improve their performance by giving them enough dataset.



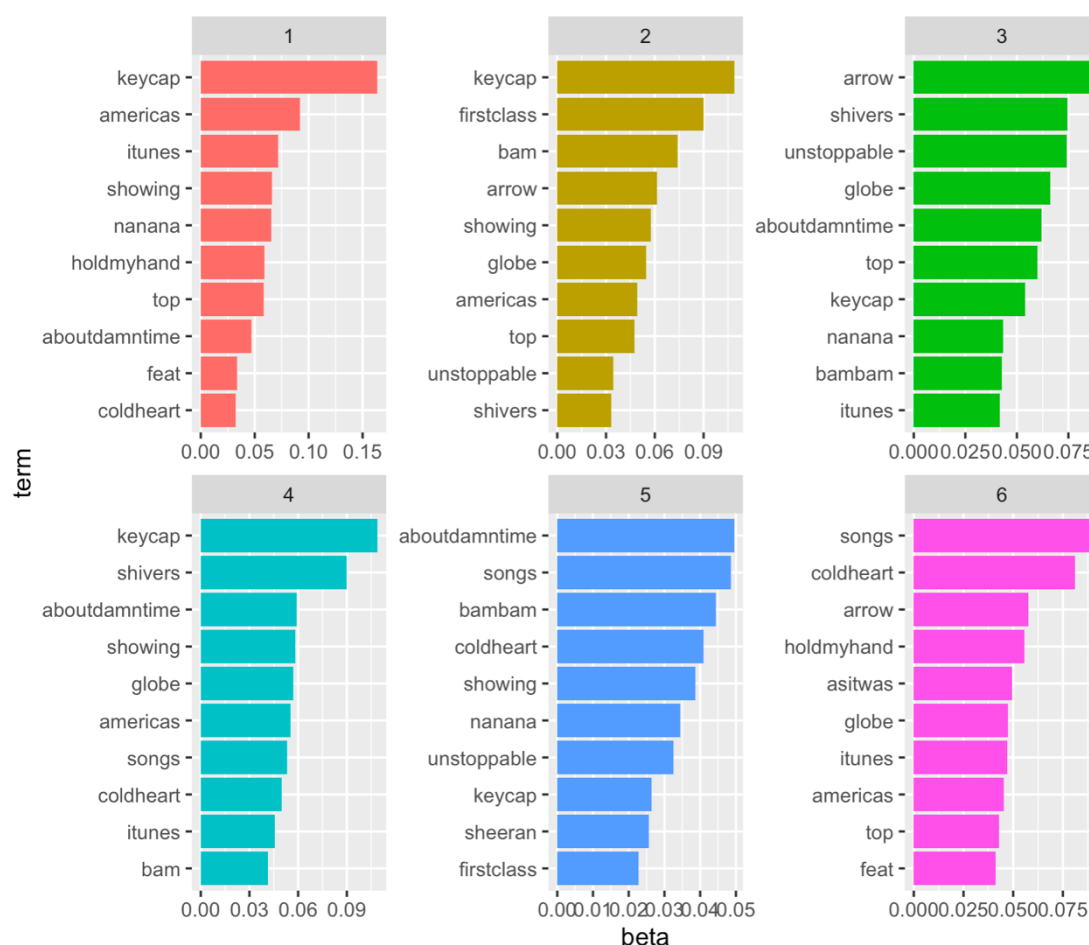
7.3 Use LDA topic modelling to identify some terms that are closely related to your artist/band. Find at least 3 significant groups of words that can be meaningful to your analysis. Explain your findings. (=> Lab 5.1)

This visualization lets us understand the six topics extracted from the dataset. Those most common in topic 5 include "sheeran", "songs", "showing", and other six song names suggesting that this topic represents Ed Sheeran's collaboration with other singers in a show.

Those most common in topics 1,3, 4 and 6 include "globe", "iTunes", "top/keycap", "americans", and other song names, suggesting that this topic represents the popular global songs on iTunes in America.

When we look at the six topics, we can find that many terms are song's name. All these topics are related to music. Either Ed Sheeran's songs or his related artists' songs.

```
ggplot(subset(word_frequ_df, freq > 2926), aes(x = reorder(word, -freq), y = freq)) +
  geom_bar(stat = "identity") +
  theme(axis.text.x = element_text(angle = 45, hjust = 1)) +
  ggtitle("Word Frequency") +
  xlab("Words") +
  ylab("Frequency")
```



8. VISUALISATION

8.1 Visualise your Twitter actor network in Gephi, with the node size determined by the number of followers for that actor. What insights can you extract from the visualisation? (This question is a little more difficult. Skip it if you're unsure and come back later.)

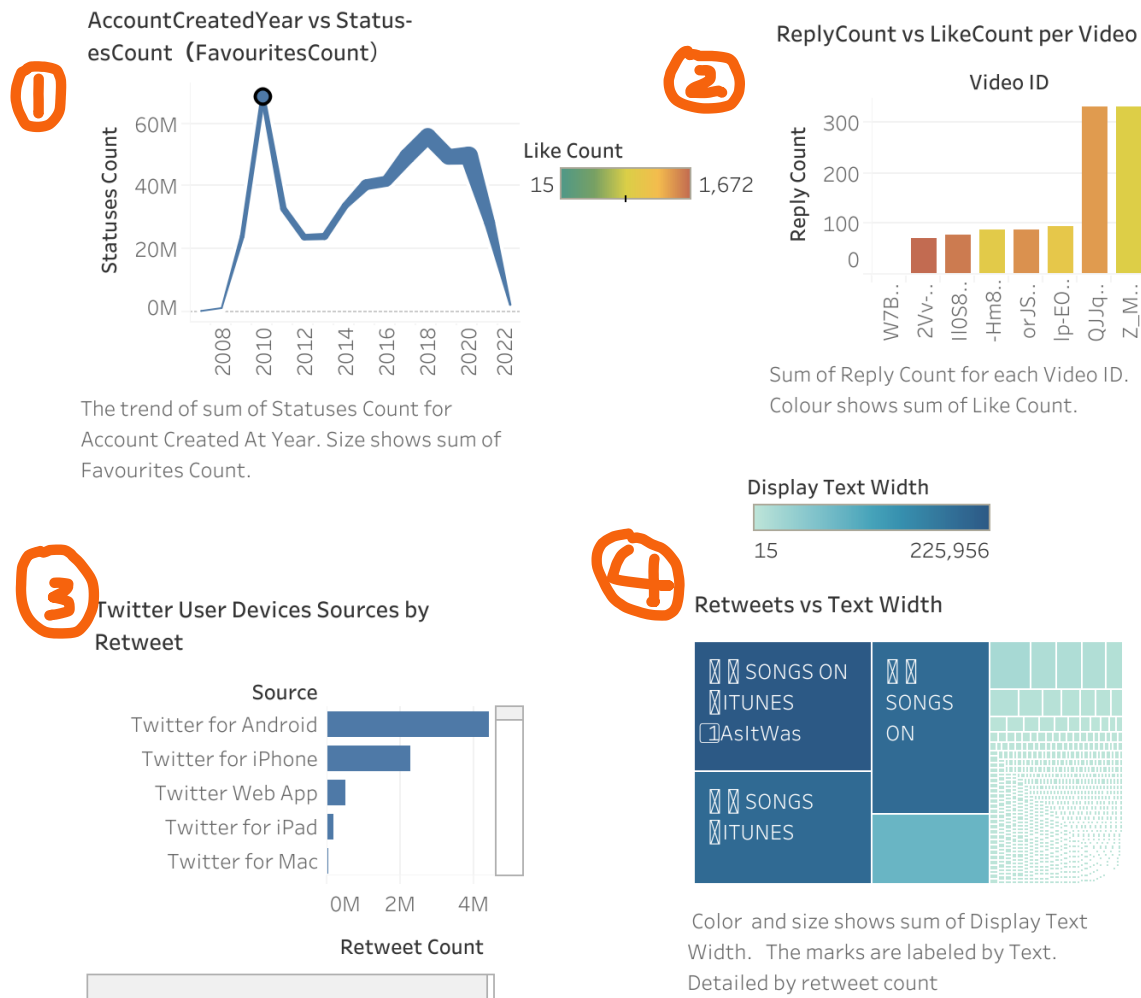
with the node size determined by the number of followers for that actor means with the node size determined by the in-degree. Comparing to 3.2, The highest followers rank similarly to the highest PageRank influencers. We can explain this that node with indegree higher than zero should all have a higher PageRank than nodes with indegree equal to zero. This also makes sense if we translate this to the idea of PageRank: nodes with indegree zero should not be considered important.

The biggest node is worldawardmusic



8.2 Create at least three charts from your datasets using Tableau and combine them together into a dashboard. Describe each chart in your dashboard and why you chose to include it. Explain the functionality of your dashboard and what insights you can obtain from it.

Ed Sheeran Social Media Analyst



	video_id	publishedAt	channelId	title
20	UAWcs5H-qgQ	2010-04-22T20:28:09Z	UC0C-w0YjGpqDXG8IHb662A	Ed Sheeran - The A Team [Official Music Video]

According to previous analysis and the above earliest published video of Ed Sheeran on YouTube platform, we get the below table.

Platforms	Foundation Year	Ed Sheeran Active Year Starting
Twitter	2006	2008
Spotify	2006	2011
YouTube	2005	2010

Chart 1:

According to the first chart from the dashboard, we know Ed Sheeran's number of active followers is distributed by their Twitter account creation year. In 2010, a large of accounts were created. At the same time, we noticed that Ed Sheeran began to publish his tracks on Spotify and YouTube. We can suggest that

- followers listen to music from Spotify and YouTube, but they prefer to share their ideas on Twitter.
- Ed Sheeran's followers created their account starting in 2016 and desire to express their likes.

Chart 2:

The data in Chart 2 comes from YouTube. The above chart shows that a high number of replies does not mean increased likes. Some videos have high engagement (high reply count), and others have higher likes. We can investigate the reasons further and make recommendations.

Chart 3:

The data in Chart 3 analyse the device distribution of retweets. As we can see from the chart, most Twitter users use android phones, followed by iPhones. Then web app and iPad. That's interesting.

Chart 4:

The data in chart 4 shows the text width and is detailed by retweets count. The exciting thing is the lengthiest text width is 225, 956. We can go further to study it.

9. ANALYSIS REVIEW

9.1 Research and review other methods/algorithms for network analysis, machine learning models, or visualisation. Compare them to the methods you used in these milestones. Did you find a method that could give you better insights or more promising results for your social media analytics? Explain why you think so.

Data source is very important. More datasets can be achieved from other music platform such as iTunes, Google music, Instagram, artist personal website and even niche music site.

We can explore Ed Sheeran's works one by one about network analysis. In this study, we analyse the whole style of the artist. We can analyse each song and each album deeply. Besides actor and semantic analysis, we can also adopt a method to analyse the diffusion of innovation.

About machine learning models: If we got more dataset, we could use neural networks algorithm to predict something more accurately. Such as predict which kind of song will be popular next month in existing listeners. Or if the incoming music event would be successful or not.

About visualization: Visualizations come in many different forms. I think we can spend more time on data exploratory because there are many variables in the Twitter dataset. Some variables we don't know the meanings of. We can show these variables in various kinds of charts, such as pie charts, maps etc., for more information we would get. That would help explain to others incoming data modelling.

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